



VICTORIA JUNIOR COLLEGE
BIOLOGY DEPARTMENT
JC2 PRELIMINARY EXAMINATIONS 2011
Higher 2

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CANDIDATE NAME

CLASS

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BIOLOGY

9648/01

Paper 1 Multiple Choice

21 September 2011

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Booklet

READ THESE INSTRUCTIONS FIRST

Write in a soft pencil.

Do not use any staples, paper clips, highlighters, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

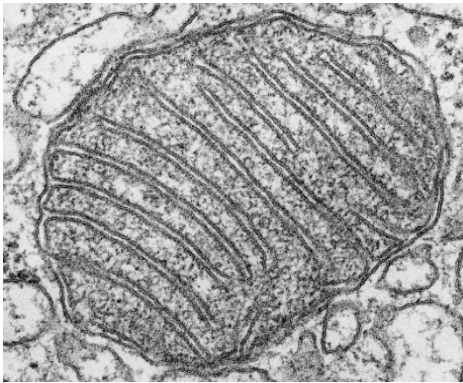
Each correct answer will score one mark. A mark will not be deduced for a wrong answer.

Any rough working should be done in this booklet

This paper consists of **22** printed pages including the cover page

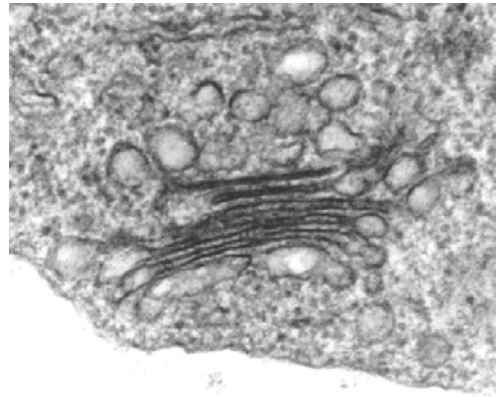
1 Which of the following eukaryotic cell structures contain nucleic acid?

I



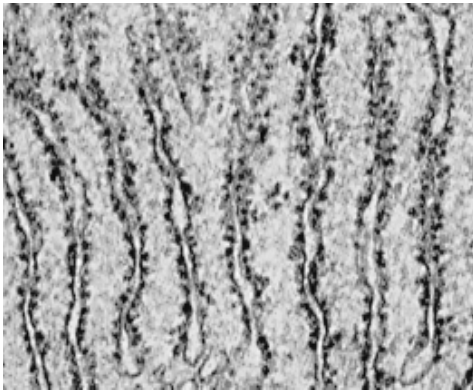
<http://antiparos.zoo.ox.ac.uk/presentations/Talks/20060516-Bioimage-DataWeb.html>

II



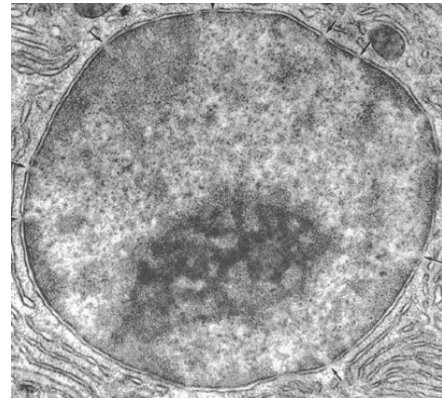
<http://course1.winona.edu/sberg/308s09/Lecture/IntracellularCompA.htm>

III



<http://bass.bio.uci.edu/~hudel/bs99a/lecture22/index.html>

IV



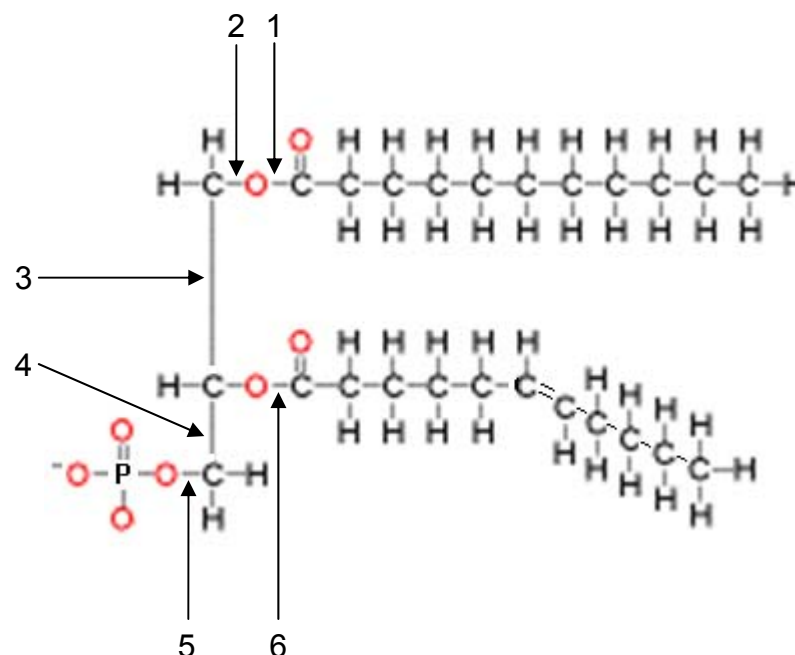
<http://course1.winona.edu/sberg/241f05/Lecture/Cells.htm>

- A IV only
- B I and IV
- C I, III and IV
- D all the above

2 Which of the following options identifies correctly the structures listed below?

	Centriole	Lysosome	Rough endoplasmic reticulum
A	It is arranged a ring of nine triplets of microtubules attached to each other by fibrils.	It releases its contents for self-digestion of a cell when the cell is damaged.	It is a double-membrane system to which ribosomes are attached.
B	It is situated next to the nucleus at a region known as centrosome.	It is formed by budding from smooth endoplasmic reticulum.	It originates from the outer membrane of the nuclear envelope.
C	It occurs as a pair at right angles to one another.	It allows increased surface area for metabolism of carbohydrate.	It provides a large surface area for attachment of mRNA.
D	It is composed of microtubules and important for organizing the spindle fibers.	It is formed by budding from Golgi apparatus.	It is a membranous system where proteins being synthesized are passed into.

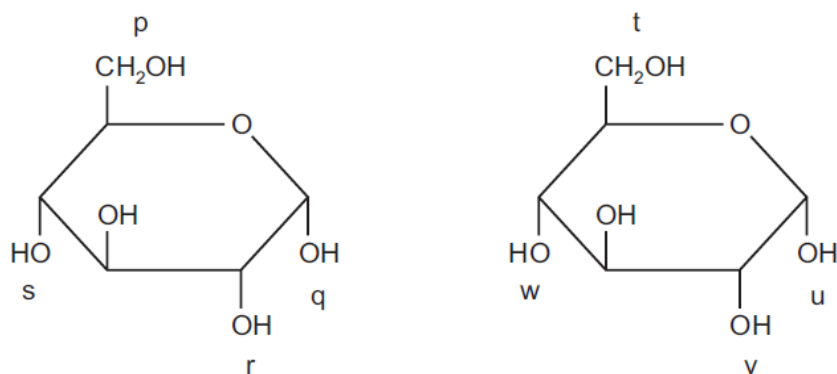
3 The diagram represents a phospholipid.



At which bonds 1 – 6 does hydrolysis occur?

- | | | | |
|----------|------------|----------|------------|
| A | 1, 3 and 5 | C | 2, 3 and 4 |
| B | 1, 5 and 6 | D | 2, 4 and 6 |

- 4 The diagram shows two molecules of glucose. Four possible bonding positions are labelled p, q, r, and s, and t, u, v, w.



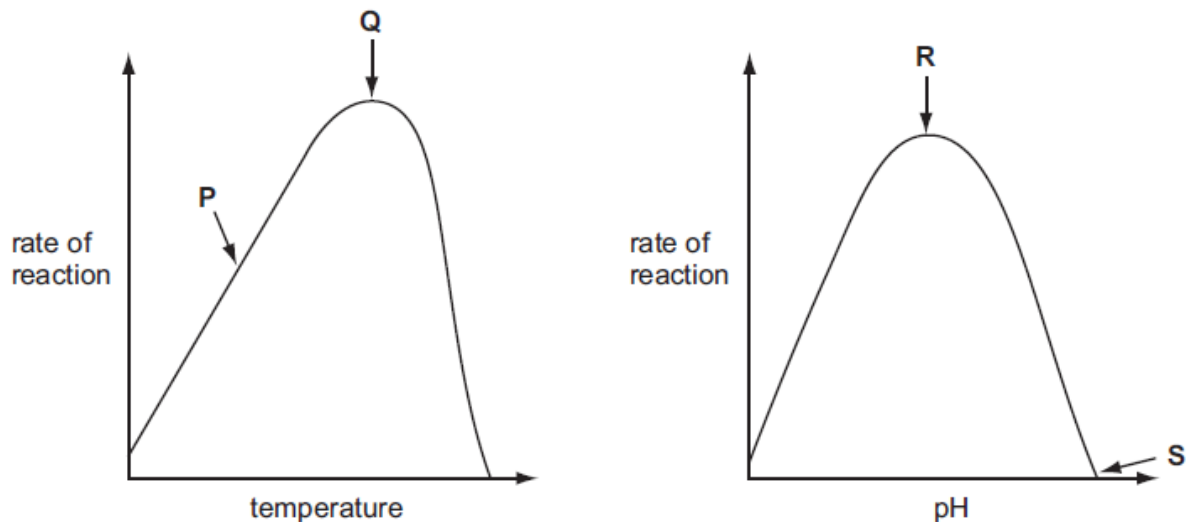
When these two molecules condense to form glycogen, where could bonds form?

- A p - u or p - v
 B p - u or q - w
 C p - v or q - w
 D p - w or v - w
- 5 Which statement describes how allosteric enzymes work?

	<i>Reversible</i>	<i>Competitive inhibition</i>	<i>End-product inhibition</i>	<i>Active and inactive forms</i>
A	×	×	✓	✓
B	✓	×	✓	✓
C	✓	✓	×	✓
D	×	✓	×	×

Key: ✓ = yes; × = no

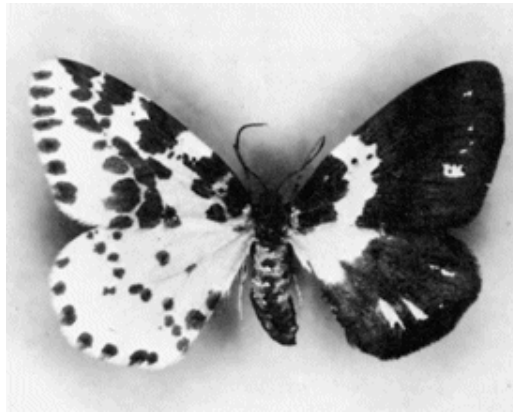
- 6 The graphs show the effects of temperature and pH on enzyme activity.



Which option explains the enzyme activity at the point shown?

	<i>P</i>	<i>Q</i>	<i>R</i>	<i>S</i>
A	The activation energy of the reaction is increasing.	The kinetic energy of enzyme and substrate is highest.	Covalent bonds are formed between enzyme and substrate.	The substrate is completely denatured.
B	Hydrogen bonds are formed between enzyme and substrate.	The number of enzyme-substrate complexes has reached a maximum.	The conformation of the active site is most ideal for binding of substrate.	Ionic bonds in the enzyme are broken and active site is lost.
C	Hydrogen bonds are formed between enzyme and substrate.	The kinetic energy of enzyme and substrate is highest.	The number of enzyme-substrate complexes has reached a maximum.	The enzyme is completely denatured.
D	The activation energy of the reaction is increasing.	Covalent bonds are formed between enzyme and substrate.	The number of denatured enzyme molecules is lowest.	Peptide bonds in the enzyme begin to break.

- 7 Mosaicism is a condition where there is more than one type of cells in the body. The picture below shows a moth, *Abraxas grossulariata*. The left side of the moth shows the typical male wing patterning, whereas the right side shows the female form.



<http://www.answers.com/topic/mosaicism>

Suggest how the condition arises?

- A Non-disjunction of the sex chromosomes during formation of the sperm
 - B Non-disjunction of the sex chromosomes during formation of the egg
 - C Non-disjunction of the sex chromosomes in mitosis in early larva development
 - D Mutation resulting in the X chromosome acquiring the genes of the Y chromosome
- 8 Which of the following describe processes that lead to an increase in variation?
- (i) breaking and rejoining in homologous chromosomes during prophase 1 of meiosis
 - (ii) random distribution of homologous chromosomes to the cell poles during anaphase 1 of meiosis
 - (iii) random variation in allele frequency with time that may result in alleles becoming more common
 - (iv) the production of new alleles by substitution of one base for another in DNA
- A (i), (iii) and (iv) only
 - B (i), (ii) and (iv) only
 - C (ii) and (iii) only
 - D (i) and (iv) only

- 9 What would be the result of analyzing part of a DNA molecule?
- A hexose sugars and phosphates in equal proportion, and an equal number of cytosine and guanine bases
 - B nucleotides and phosphates in equal proportion, and an equal number of adenine and cytosine bases
 - C pentose sugars and phosphates in equal proportion, and an equal number of adenine and thymine bases
 - D twice as many phosphates as pentose sugars, and an equal number of adenine and guanine bases
- 10 Which of the following statements does/do **not** apply to DNA replication in both bacteria and eukaryotes?
1. Replication begins at a single origin of replication.
 2. Replication proceeds in a 5' to 3' direction only and is bidirectional from the origin(s).
 3. Replication occurs continuously for both strands of new DNA in bacteria since it is a circular DNA.
 4. There are numerous different bacterial chromosomes, with replication occurring in each at the same time.
- A 2 only
 - B 2 and 3
 - C 1 and 4
 - D 1, 3 and 4
- 11 Which of the following statements about RNA is **incorrect**?
- A Small nuclear RNAs bind with ribonucleoproteins to form spliceosomes.
 - B Ribosomal RNAs bind with transfer RNA to catalyse the formation of peptide bonds.
 - C Transfer RNA translates the base sequence present on the DNA into the amino acid sequence in a protein.
 - D The base sequences in the RNA molecules are complementary to the base sequences in the DNA that is being transcribed.

12 The sequence of DNA bases

GGCAATTGGAAACGAATACCCAGT

codes for the following sequence of amino acids.

glycine- asparagine-tryptophan-lysine-arginine-isoleucine-proline-serine

A mutant organism, in which one of the bases was deleted and then inserted at a different point in the DNA molecule, produced a peptide with the following sequence:

glycine- asparagine-tryptophan- asparagine -glutamine-isoleucine-proline-serine

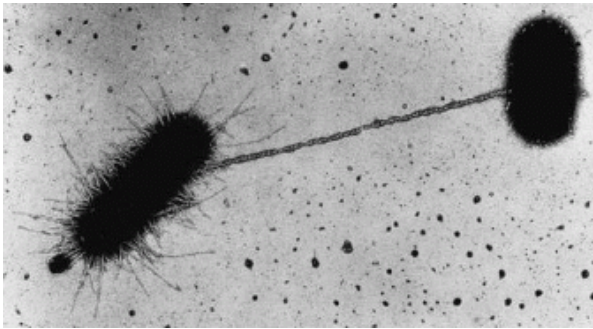
Which base was removed by this mutation?

- A** adenine
- B** guanine
- C** cytosine
- D** thymine

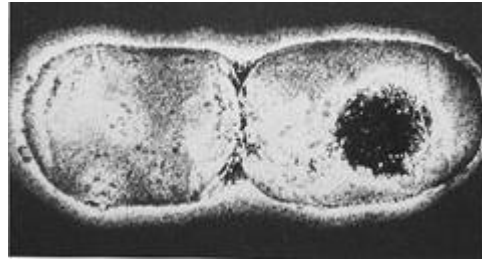
13 Why are antibiotics ineffective against viruses?

- A** Viruses mutate quickly when challenged by an antibiotic.
- B** Viruses have developed resistance to antibiotics.
- C** Viruses destroy T-lymphocytes before the antibiotic can work.
- D** Viruses do not have metabolic pathways for the antibiotic to target.

- 14** The photomicrographs below show two different processes occurring in two different species of bacteria.



Process 1



Process 2

Which of the following statements is/are true of both processes?

- (i) For both processes, only bacteria with genes that code for cytoplasmic bridge are involved.
- (ii) Process 1 requires direct contact between 2 different bacteria whereas process 2 can occur with 1 bacterium.
- (iii) Process 1 will result in an increase in the number of identical bacteria whereas process 2 will result in an increase in the number of different bacteria.
- (iv) Both processes involve DNA replication.

- A** (i) and (iii) only
- B** (ii) and (iv) only
- C** (i), (ii) and (iv) only
- D** (i), (ii), (iii) and (iv)

- 15** Which of the following statements could explain why a combination of different drugs rather than a single drug is being used to treat HIV patients?
- (i) HIV has a short generation span (eg. 2 days).
 - (ii) In an AIDS patient, the HIV infection produces many new viruses per day (eg. 10^{10} new viruses per day or more).
 - (iii) HIV has an RNA genome which has a higher mutation rate than DNA as it is single stranded.
 - (iv) The RNA genome allows the HIV to mutate rapidly to acquire the resistance to the drug being used.
 - (v) Insertion of the HIV genome into the host cells will result in the host cells mutations that will confer resistance to the drug.
- A** (i) and (ii) only
B (iii) and (v) only
C (i),(ii), and (iii) only
D (iii), (iv) and (v) only
- 16** In the immune system, B cells are lymphocytes that can produce different antibodies that target foreign agents that invade the body.
- Which of the following provides a possible mechanism for the production of the different antibodies in a B cell?
- A** Crossing over and random segregation to create new combination of alleles
B Acetylation of histone proteins to create different levels of packaging in chromatin model so as to allow for different genes to be expressed
C Activation of different specific transcription factors to regulate gene expression
D Splicing of selected alternative exons into final mRNA products
- 17** Which of the following mechanism is needed before the transcription of a gene can begin?
- A** Removal of acetyl groups from specific lysine residues in histones
B Binding of snRNP to specific sequences on the DNA to form a complex
C Attachment of transcription factors onto promoter site upstream of the gene
D Removal of methyl groups from the DNA

- 18** Which one of the following statements correctly describes the control of transcription for genes encoding for enzymes in a metabolic pathway?
- A** The genes are transcribed as a single transcription unit, with each gene having its own promoter.
 - B** The genes respond equally to the same set of general transcription factors in a complex, but each individual gene will respond differently to certain specific transcription factors.
 - C** The genes have various combinations of control elements that enable different activators and repressors to bind and affect the rate of transcription.
 - D** The genes are found close to each other on the chromosome with chromatin remodelling as a form of transcriptional control.
- 19** Which of the following observations does **not** support the development of cancer as a multistep process?
- A** Cancer involves the accumulation of mutations in at least one oncogene and several tumor suppressor genes in a specific order.
 - B** The lag time that often separates exposure to a cancer-causing agent and the development of cancer.
 - C** An increase risk of people who inherit predisposing mutations to developing cancer.
 - D** The incidence of cancer increases with age.

20 Three gene loci in mice and the characteristics they control in mice are shown below.

Locus 1 (A/a) determines coat colour	Locus 2 (K/k) Determines tail shape	Locus 3 (F/f) Determines condition of coat
agouti vs albino	kinky vs straight	frizzy vs non-frizzy

A student studying the inheritance of these characteristics carried out controlled crosses involving two loci at any one time. The outcomes of his experiments are shown below.

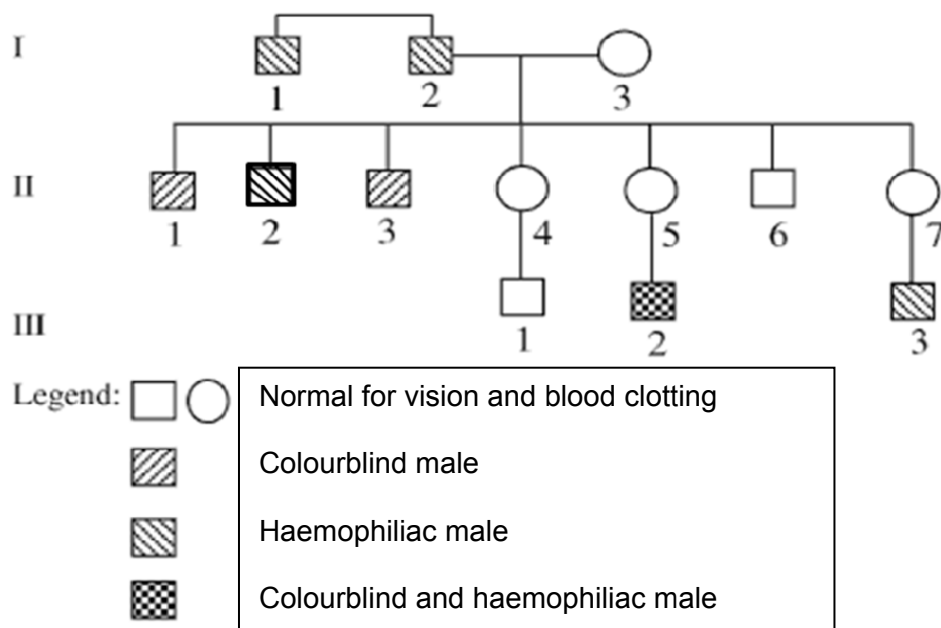
Parents (pure bred)	F1	Offspring of a test cross with F1
<i>Cross 1</i> agouti, non-frizzy coat X albino, frizzy coat	agouti, non-frizzy coat	agouti, non-frizzy 44 albino, frizzy 46 agouti, frizzy 5 albino, non-frizzy 5
<i>Cross 2</i> agouti, straight tail X albino, kinky tail	agouti, kinky tail	agouti, straight 23 albino, kinky 27 agouti, kinky 24 albino, straight 26

Which one of the following is a valid conclusion from his results?

- A** The genotype of the F1 of cross 1 is $\frac{A\ f}{a\ F}$
- B** The map distance between albino and non-frizzy is 5 map units
- C** The gene for coat colour and tail shape assort independently.
- D** Since there are more recombinants formed in cross 2, one possible arrangement of the 3 genes is agouti, non-frizzy, kinky.

For questions 21 and 22, refer to the pedigree below.

21 The pedigree below traces the inheritance of 2 sex-linked genetic diseases in a family.



Which of the following options regarding the inheritance is false?

You are to assume that the fathers of generation III males are normal for colour vision as well as blood clotting.

- A** Crossing over between the 2 genes has occurred during gamete formation in individuals I-3 and II-5.
- B** Individual III-2 has gotten the allele for haemophilia from his grandfather I-2.
- C** The genotype of II-5 can be either $X_h^C X_H^c$ or $X_H^C X_h^c$
- D** The genotype of I-3 is $X_h^C X_H^c$

22 If individual 5 were to have another child, what is the probability of this child being a colourblind, assuming no crossovers occur.

- A** 0
- B** 1/8
- C** 1/4
- D** 1/2

- 23** Purple buds of the morning glory flower, *Ipomoea*, open into blue flowers. As the flower opens, the pH on the vacuoles of the flower epidermal cells increases and this results in a change of colour from purple to blue.

A mutant purple-flowered morning glory plant carries recessive alleles of a gene **B/b**, coding for a membrane-bound ion pump, and is unable to increase the pH of the vacuole.

Both normal blue flowers and mutant purple flowers have the same anthocyanin pigment, coded by the dominant allele of the gene **A/a**. Plants with **aa** cannot produce anthocyanin and they have white flowers.

The genes **A/a** and **B/b** are not linked.

A blue-flowered morning glory plant was crossed with a purple-flowered plant. Their offspring consisted of plants which are blue-flowered, purple-flowered as well as white-flowered.

What were the genotypes of the blue-flowered and purple-flowered plants?

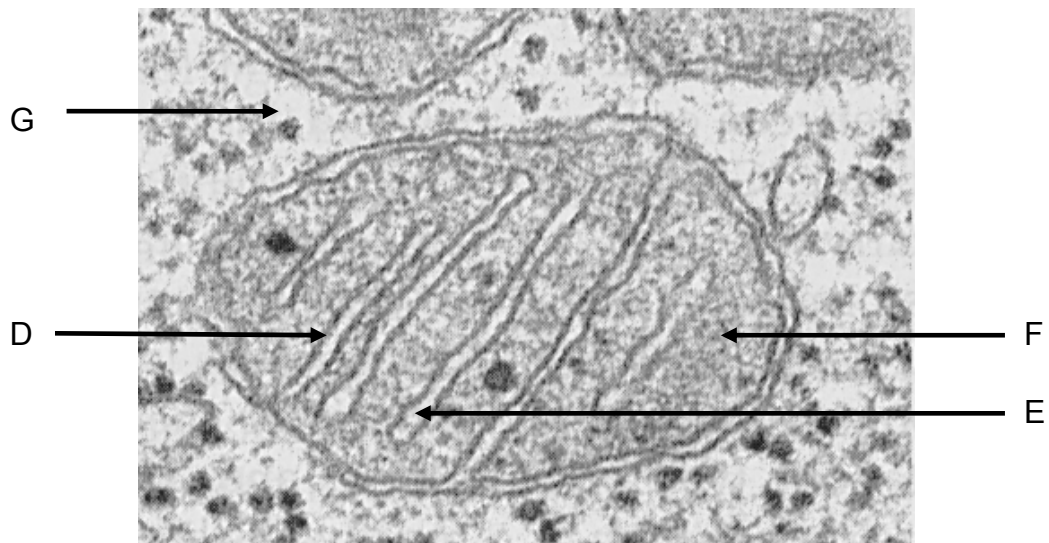
	<i>Blue-flowered plant</i>	<i>Purple-flowered plant</i>
A	AABB	AaBb
B	AaBb	Aabb
C	aaBB	AaBB
D	aaBb	aaBb

- 24** Which one of the following substances, when added, would directly result in a decline in ATP production in glycolysis?

- (i) A chemical that would bind to NAD^+ irreversibly and induces its reduction to NADH.
- (ii) An inhibitor that has a similar structure to glucose but cannot be broken down by respiratory enzymes.
- (iii) A chemical that creates an anaerobic environment by combusting in oxygen
- (iv) A reagent that binds to the active site of ATPase permanently

- A** (i) and (ii) only
- B** (iii) and (iv) only
- C** (i), (ii) and (iv) only
- D** All of the above

- 25 Match the following processes with structures labeled D–G in the electron micrograph below.



<http://www.cbs.dtu.dk/staff/dave/roanoke/bio101ch05.htm>

	<i>Drop in pH observed</i>	<i>Attachment of inorganic phosphate to ADP</i>	<i>Series of reduction / oxidation reactions</i>	<i>Carbon dioxide evolved</i>
A	D	E	F	G
B	E	G	D	F
C	G	F	D	E
D	F	G	E	D

- 26 A metabolic poison rotenone was added to a mixture of homogenized cells with intact chloroplasts. It is discovered that rotenone can bind irreversibly to the electrons carriers on the thylakoid membrane in chloroplasts and alter their conformational shape.

In what way would the Calvin cycle be affected by rotenone poisoning?

- A** Calvin cycle will stop, as the accumulation of triose phosphate will create feedback inhibition on glucose formation.
- B** The rate of Calvin cycle will decrease due to the accumulation of glycerate-3-phosphate.
- C** The rate of Calvin cycle will increase since electrons can flow directly to final electron acceptor to form reduced NADPH.
- D** There will be no effect since Calvin cycle is an enzyme-controlled process.

27 Which of the following statements is true regarding cyclic and non-cyclic photophosphorylation?

- A** Cyclic photophosphorylation only requires light at 680nm to function optimally while non-cyclic photophosphorylation requires light at both 680nm and 700nm to function optimally.
- B** Cyclic photophosphorylation is dependent on products generated through non-cyclic photophosphorylation.
- C** Only non-cyclic photophosphorylation utilizes electrochemical gradient generated to drive ATP production.
- D** Only cyclic photophosphorylation can function in the absence of photosystem II.

28 Which of the following is **not** a consequence of increased cellular levels of cAMP?

- A** Activation of a kinase cascade
- B** Activation of adenylyl cyclase
- C** Increased phosphorylation of glycogen phosphorylase
- D** Inhibition of glycogen synthesis

29 Which of the following process(s) leads to signal amplification in the cell?

- (i) Binding of multiple signal molecules to the membrane bound receptor
- (ii) Phosphorylation of the first kinase of a cascade
- (iii) Branching of different signalling pathways that produces multiple cellular responses
- (iv) Activation of numerous intracellular receptors that acts as transcription factors in gene expression

- A** (ii) only
- B** (ii) and (iii) only
- C** (i) and (iv) only
- D** All of the above

- 30** Certain drugs act at synapses and affect the action of neurotransmitter substances. The table shows the effects of four developmental drugs.

Drug	Effect
I	Fits snugly into the active site of cholinesterase
II	Induce the opening of voltage-gated Na^+ ion channels
III	Inhibits transfer of ions through the Na^+/K^+ pump
IV	Promotes the activity of proteins that allow Ca^{2+} ions into the pre-synaptic terminal

Which of the drug(s) would increase the chances of generating an inhibitory postsynaptic potential at the junction of an inhibitory synapse?

- A I and IV only
 - B II and III only
 - C II and IV only
 - D I, II and III
- 31** The decline in sickle cell anaemia among North Americans of African origin is thought to be due to an absence of selection pressure by malaria.

Which other factor has had the most influence in this decline?

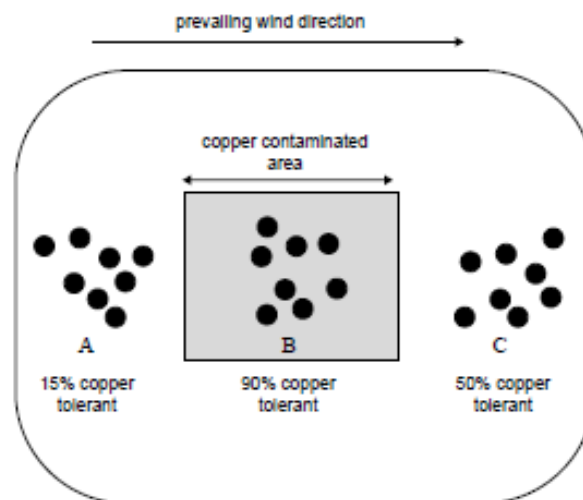
- A Outbreeding with other races
- B Inbreeding within their community
- C New mutations to the sickle cell allele
- D Better drugs to combat sickle cell anaemia

32 Which of the following statements is/are true of genetic drift?

- (i) Genetic drift requires the presence of variation.
- (ii) Genetic drift can result in the loss of beneficial alleles.
- (iii) Founders effect and genetic bottleneck drive genetic drift
- (iv) Genetic drift occurs in small population only

- A** (i) and (iv) only
- B** (ii) and (iii) only
- C** (i), (ii) and (iii) only
- D** (ii), (iii) and (iv) only

33 Figure below shows an aerial view of a region which included a section of copper contaminated soil (shaded grey on the diagram). The black circles represent populations of grass growing in areas A, B and C. The diagram also includes the percentage of copper resistant plants in areas A, B and C. Tolerance to copper in the soil is genetically determined.



It is reasonable to conclude from the information that

- A** all plants in the copper contaminated area would be homozygous at the gene locus for copper tolerance.
- B** high levels of copper tolerance would be a selective advantage for all plants in areas A, B and C.
- C** the difference in copper tolerance between populations in areas A and C is the result of mutation.
- D** gene flow is occurring between populations in areas B and C.

- 34** A botanist identifies a new species of plant that has 32 chromosomes. It grows in the same habitat with three similar species: species A ($2n = 14$), species B ($2n = 16$) and species C ($2n = 18$) and is fertile.

What is the most possible mode of speciation mechanism for the new species?

- A** Allopolyploid, with hybrid that is formed from fusion of the gametes from species A and C undergoing non-disjunction during early development.
 - B** Allopolyploid with a hybrid being formed from the fusion of gametes of species A and B that have undergone independent non-disjunction.
 - C** Autopolyploidy with a hybrid being formed from the non-disjunction of the chromosomes during mitosis in B.
 - D** Allopatric speciation by development of extra chromosomes as a reproductive isolating mechanism.
- 35** Probes are short, single nucleotide strands that are used in identification of DNA fragments with a specific sequence.

Which of the following statements about probes is false?

- A** Probes have to be labelled in order to be visualized.
 - B** A probe isolated from one organism can be used to locate a homologous DNA fragment from another unrelated organism.
 - C** They must be produced with the same restriction endonuclease as the DNA fragments.
 - D** Shorter probes are able to adhere to more fragments compared to longer probes.
- 36** An exploratory gene treatment was used to treat an autosomal recessive disease Phenylketonuria. The dominant allele was introduced into patients via a retroviral-mediated delivery system. Observations made after treatment showed no signs of improvement on the patients.

Which of the following would **not** be a reasonable explanation behind the failure of this treatment?

- A** An immune response was elicited against the vector.
- B** The expressed protein was unable to be folded into the correct conformation.
- C** A single copy of the allele results in inadequate expression of proteins to effect upon the entire organism.
- D** The genetic sequence was integrated into a silencer region on the chromosome.

- 37** A sample of DNA was treated separately with two different restriction enzymes. The products of digestion was then run on a gel and stained with ethidium bromide.

Which of the following statement(s) **cannot** account for the difference in the *number* of bands seen in the gel?

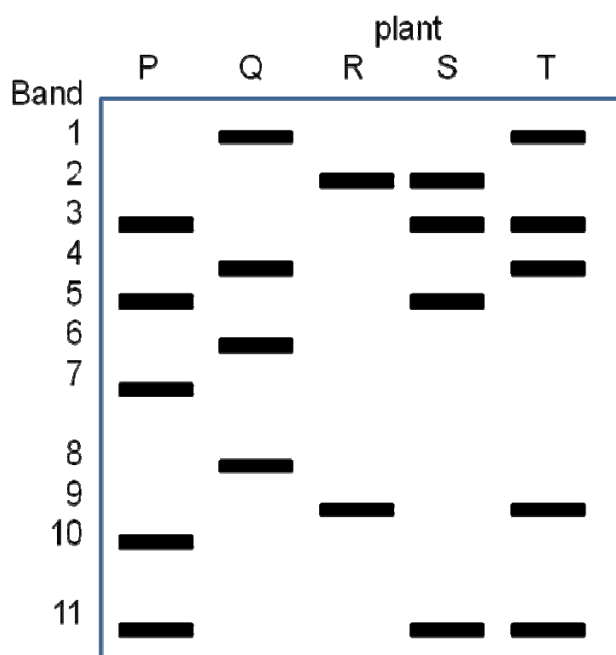
- (i) The two enzymes have active sites that are of different shapes.
- (ii) One of the enzymes produces blunt ends and the other sticky ends.
- (iii) The position of the restriction sites for each restriction enzyme differs
- (iv) Ethidium bromide binds specifically to only certain DNA fragments.

- A** (ii), (iii) and (iv) only
- B** (i) and (iii) only
- C** (ii) and (iv) only
- D** (ii) only

- 38 A possible route of escape of an inserted gene from a genetically engineered crop of oil-seed rape, *Brassica napus*, is into populations of the wild turnip, *Brassica rapa*. Two populations of wild turnip growing next to large fields of a cultivar of oil-seed rape were studied. Seeds were collected from these wild turnips plants and their DNA analysed using restriction enzymes and electrophoresis.

Using the key provided below, determine

- whether plants S and/or T are interspecific hybrids and
- the parental plants they are derived from



Key
P oil-seedrape
Q wild turnip plant 1
R wild turnip plant 2
S & T possible interspecific hybrid

	<i>Interspecific hybrid</i>	<i>Parent plants</i>
A	Both S and T	S: P x R T: P x Q
B	Both S and T	S: P x Q T: P x R
C	S only	P x R
D	T only	Q x R

- 39 Which of the following is **not** a concern with growing genetically engineered crops eg. *Bt* corn?
- A Flow of the Bt gene from cultivated corn to wild relative thereby enhancing the weediness of certain corn relatives, which could then lead to loss of biodiversity
 - B Transfer of marker genes that convey antibiotic resistance, used in the identification of successfully transformed cells, to bacteria that resides in the gut of humans thereby rendering the bacteria antibiotic resistant.
 - C Toxic effects of Bt on non-target lepidopteran insects eg. monarch butterfly larvae (of the insect order *Lepidoptera*) could upset ecological balance.
 - D Transfer of Bt genes to the pests thereby increasing their resistance to the Bt toxin
- 40 Which of the following regarding embryonic stem cells and hematopoietic stem cells is true?
- A As embryonic stem cells develop, they turned into hematopoietic stem cells as they lose their ability to differentiate into all types of cells.
 - B Embryonic stem cells have more genes than hematopoietic stem cells and thus are able to form more types of cells.
 - C Under normal conditions, embryonic stem cells express more of their genes compared to the hematopoietic stem cells.
 - D Both stem cells are derived from the zygotic stem cells with the hematopoietic stem cells having a lowered telomerase activity compared to the embryonic stem cells.